

E-COMMERCE CONVERSION DYNAMICS: CUSTOMER BEHAVIOR & PURCHASE INTENT ANALYSIS

Customer Behavior, Conversion
Dynamics & Purchase Intent



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly &
ML libraries

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Overview & Business Objective



1.1 PROJECT OVERVIEW

E-commerce platforms operate in a dynamic environment where purchase decisions are influenced by product engagement, traffic sources, visitor type & session behavior.

Understanding these behavioral patterns helps businesses identify sessions associated with stronger purchase conversion and better understand online customer activity.

The E-Commerce Conversion Dynamics project uses session-level online shopping data and machine learning to analyze purchase behavior and predict whether a session results in a completed purchase.

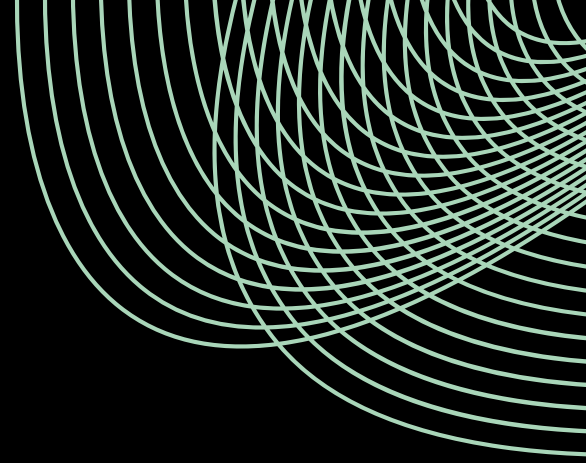
The project combines:

- Data understanding and cleaning
- Exploratory Data Analysis
- Behavioral feature engineering
- Class imbalance handling
- Machine learning classification
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented insights

The objective is not only to predict purchase conversion accurately, but also to understand the behavioral signals behind the model's predictions and how these insights can support better e-commerce decision-making.



1.2 PROBLEM STATEMENT



Purchase conversion is influenced by multiple interacting aspects of an online shopping session, making it difficult to identify potential purchases using individual behavioral indicators alone.

The key problem addressed in this project is:

How can session-level e-commerce data be used to build a machine learning model that accurately identifies purchase sessions while also understanding the key behavioral factors associated with conversion?

The project therefore treats **Revenue** as the target variable and uses browsing behavior, product engagement, page value, traffic sources, visitor characteristics, and technical session information as predictive features.



1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Understanding the structure, quality, and characteristics of the online shopping dataset.
- Analyzing purchase conversion patterns across months, visitor types, traffic sources, and session behavior.
- Identifying relationships between browsing activity, product engagement, and purchase outcomes.
- Creating meaningful behavioral features and preparing the dataset for machine learning.
- Addressing class imbalance and developing multiple classification models for purchase prediction.
- Tuning and comparing classification models using appropriate performance metrics.
- Evaluating classification thresholds and investigating false-positive and false-negative predictions.
- Explaining model predictions using feature importance and permutation importance.
- Translating analytical and modeling results into practical e-commerce business insights.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, behavioral feature engineering, machine learning, model evaluation, and explainability to understand purchase patterns, identify key conversion signals, evaluate predictive performance, and derive actionable e-commerce insights.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Purchase conversion patterns
- Behavioral relationship analysis

02

Conversion Modeling

- Feature preparation
- Behavioral feature engineering
- Classification modeling
- Model tuning and comparison

03

Explainability

- Feature importance analysis
- Original-feature aggregation
- Permutation importance analysis

04

Business Interpretation

- Purchase conversion observations
- Customer behavior insights
- Conversion signal interpretation
- Recommendations for e-commerce optimization



Data Understanding & Exploratory Data Analysis



2.1 DATA OVERVIEW

The dataset provides a comprehensive view of e-commerce shopping sessions, covering browsing activity, page engagement, visitor characteristics, technical attributes, traffic sources, special-day proximity, and purchase outcomes. These variables provide the foundation for analyzing customer behavior, identifying factors associated with purchase conversion & developing predictive models for e-commerce purchase intention.

Data Source: UCI

Dataset Size

12,330

Records

18

Original Features

Purchase Diversity

9

Regions

3

Visitor Type

20

Traffic Type



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Administrative	Numerical (Discrete)	Number of administrative pages visited.
Administrative_Duration	Numerical (Continuous)	Time spent on administrative pages.
Informational	Numerical (Discrete)	Number of informational pages visited.
Informational_Duration	Numerical (Continuous)	Time spent on informational pages.
ProductRelated	Numerical (Discrete)	Number of product-related pages visited.
ProductRelated_Duration	Numerical (Continuous)	Time spent on product-related pages.
BounceRates	Numerical (Continuous)	Average bounce rate during the session.
ExitRates	Numerical (Continuous)	Average exit rate during the session.
PageValues	Numerical (Continuous)	Average value of pages visited.
SpecialDay	Numerical (Continuous)	Proximity to a special shopping day.
Month	Categorical	Month of the shopping session.
OperatingSystems	Categorical	Operating system used by the visitor.
Browser	Categorical	Browser used by the visitor.
Region	Categorical	Geographic region of the visitor.
TrafficType	TrafficType	Type of traffic source.
VisitorType	Categorical	Type of visitor.

DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Weekend	Categorical (Binary)	Indicates whether the session was on a weekend.
Revenue	Categorical (Binary)	Indicates whether a purchase was completed.
Product ID	Categorical	Unique identifier of the product
Category	Categorical	Product category
Region	Categorical	Geographic region of the store
Inventory Level	Numerical (Discrete)	Inventory available at the time of observation
Units Sold	Numerical (Discrete)	Number of units sold
Units Ordered	Numerical (Discrete)	Number of units ordered
Price	Numerical (Continuous)	Selling price of the product
Discount	Numerical (Discrete)	Discount applied to the product
Weather Condition	Categorical	Weather condition during the observation
Promotion	Binary	Indicates whether a promotion was active
Competitor Pricing	Numerical (Continuous)	Price of the comparable product offered by competitors
Seasonality	Categorical	Seasonal classification of the observation
Epidemic	Binary	Indicates whether an epidemic-related condition was present
Demand	Numerical (Discrete)	Target variable representing product demand

2.2 TARGET VARIABLE

The target variable selected for prediction is:

Revenue

It represents whether the session resulted in a completed purchase.

True: *Purchase*

False: *No Purchase*



The target is imbalanced, with purchase sessions representing a smaller proportion of the overall dataset. This imbalance was therefore considered during model development.



2.3 DATA QUALITY & PREPARATION OVERVIEW

Before modeling, the dataset was examined for:

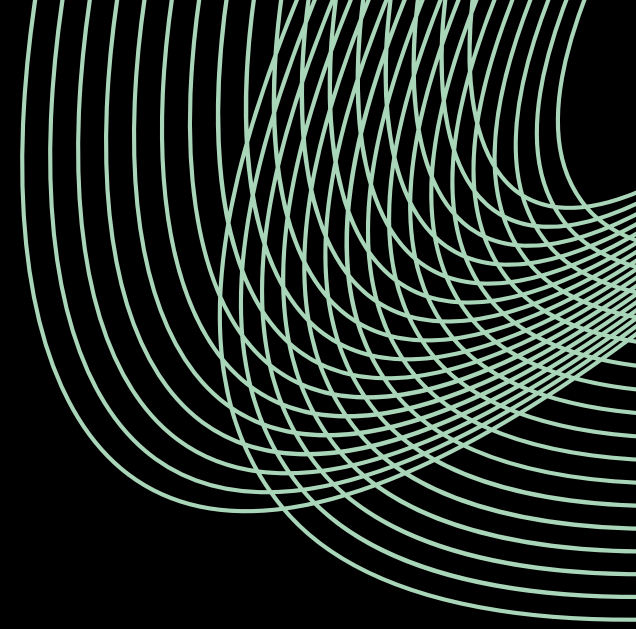
- Data types
- Missing values
- Duplicate observations
- Categorical variables
- Numerical variables
- Target variable distribution
- Potential outliers
- Feature relationships
- Class imbalance

Duplicate observations were removed, categorical variables were prepared for machine learning using appropriate encoding, and numerical features were assessed for consistency

Additional behavioral features were engineered from page visits and browsing duration to capture overall session engagement.

The target variable was also examined for class imbalance, with SMOTE applied during model training to address the minority purchase class.





2.4 Exploratory Data Analysis



New visitors show the highest purchase conversion rate, while returning visitors record the lowest.

\$2,135

Average Cost per Service



Key observations

- New visitors have the highest conversion rate at 24.91%, followed by the Other visitor group at 18.82%.
- Returning visitors have the lowest conversion rate among the three groups at 13.93%.

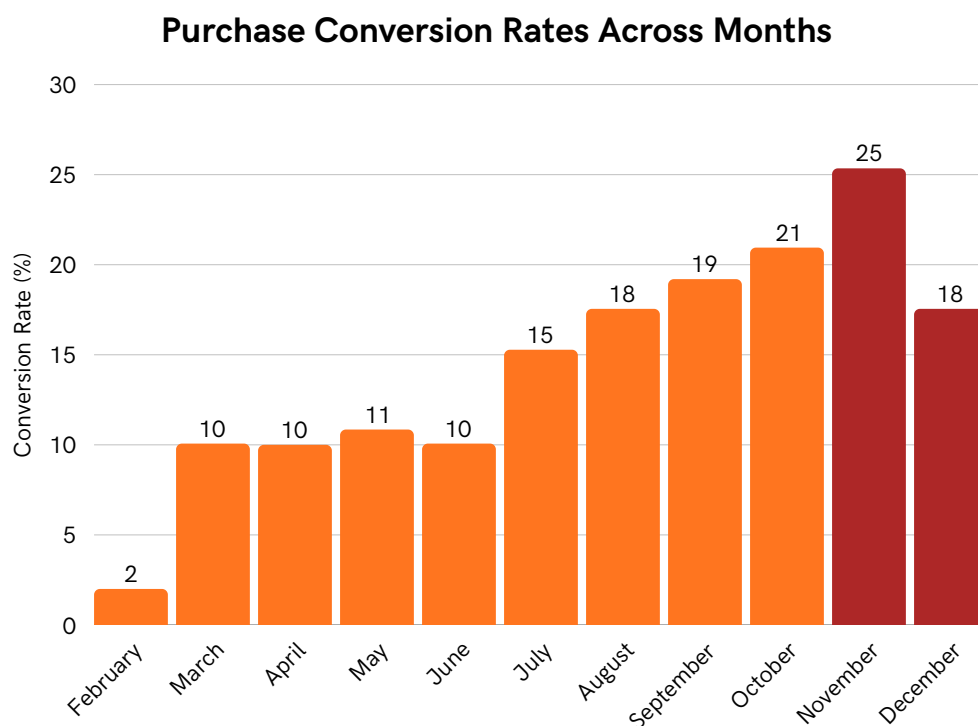
Business Insights

- Examine the browsing and acquisition characteristics of new visitors to understand the behaviors associated with their higher conversion rate.
- Investigate the lower conversion rate among returning visitors to identify potential opportunities to improve repeat-session conversion.

Purchase conversion rates vary substantially across months, with November showing the highest conversion rate.

\$1,251

Typical Cost per Resource (Median)



Key observations

- November records the highest conversion rate at 25.35%, followed by October at 20.95%.
- February records the lowest conversion rate at 1.63%, while March to June records around 10%.

Business Insights

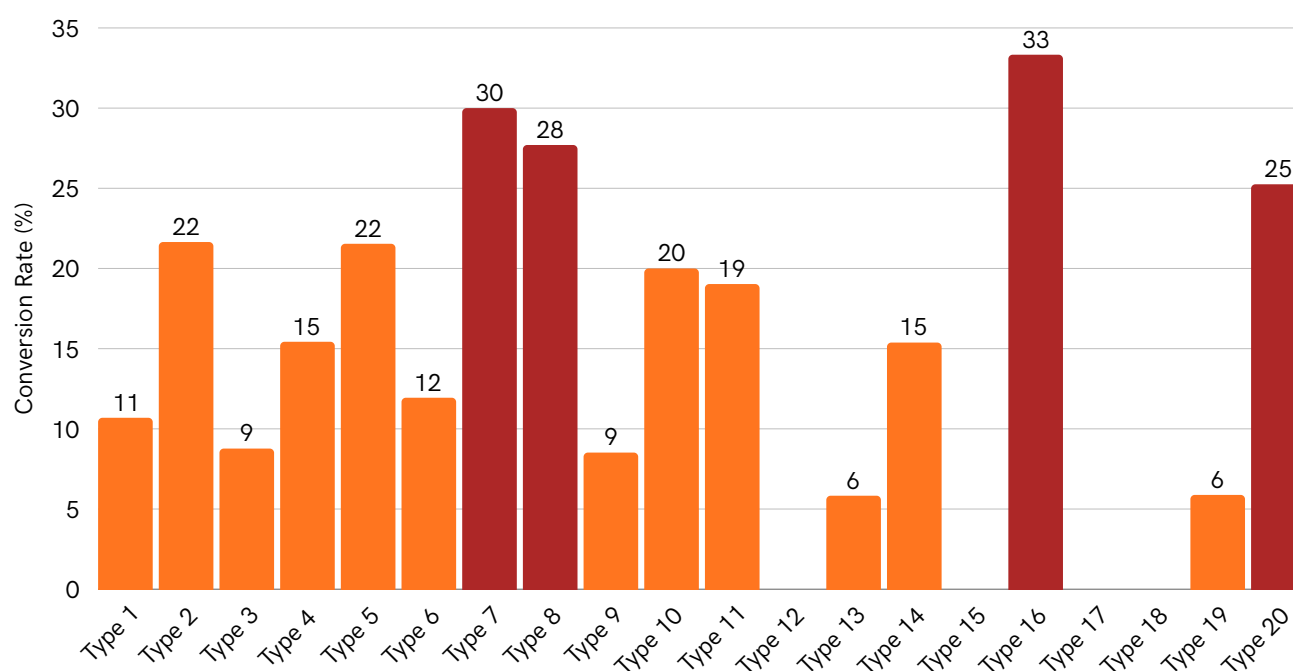
- Prioritize conversion-focused campaigns and promotional activity during months showing stronger conversion rates, particularly around October and November.
- Investigate the behavioral and commercial factors associated with low-conversion months such as February to identify opportunities for improvement.

Purchase conversion rates vary considerably across traffic sources, with a small number of traffic types showing substantially higher rates.

\$2,147

Average Spending per Region

Purchase Conversion Rates Across Traffic Type



Key observations

- Traffic Type 16 records the highest conversion rate at 33.33%, followed by Traffic Type 7 at 30.00% and Traffic Type 8 at 27.70%.
- Several traffic types record very low or 0% conversion rates, while Traffic Type 20 reaches 25.25%.

Business Insights

- Evaluate high-conversion traffic sources to understand which acquisition channels are associated with stronger purchase conversion.
- Review low- and zero-conversion traffic sources to identify opportunities for traffic-quality improvement or more targeted acquisition strategies.

Purchase sessions generally show higher product-related page engagement than non-purchase sessions.



Key observations

- The median number of ProductRelated pages is higher for purchase sessions than for non-purchase sessions.
- Both groups contain substantial high-engagement outliers, with non-purchase sessions reaching above 700 product-related page visits and purchase sessions reaching above 500.

Business Insights

- Product-page engagement can be monitored as an important behavioral signal when identifying sessions with stronger purchase intent.
- High product-page activity does not guarantee conversion, so additional behavioral signals should be considered when distinguishing purchasers from highly engaged non-purchasers.

Product-related engagement & browsing duration show a positive relationship, with purchase and non-purchase sessions distributed across the same overall engagement range.



Key observations

- Product-related pages visited and product-related page duration increase together, indicating that sessions with more product-page activity generally spend more time on those pages.
- Purchase and non-purchase sessions substantially overlap across the engagement range, although purchase sessions are visible throughout higher-engagement regions.

Business Insights

- Combine product-page engagement and browsing duration with other conversion signals when assessing session-level purchase intent.
- Investigate high-engagement non-purchase sessions to understand where substantial product interest does not translate into a recorded purchase.

Purchase sessions show substantially higher Page Value levels than non-purchase sessions

52.03%

Typical CPU Utilization (Median)



Key observations

- The distribution for purchase sessions is clearly shifted upward, with a much higher central range than non-purchase sessions.
- Non-purchase sessions are heavily concentrated near zero Page Value, while purchase sessions extend across substantially higher values.

Business Insights

- Page Value can serve as a strong behavioral signal when identifying sessions with stronger purchase potential.
- Monitor Page Value alongside other session characteristics rather than treating it as a standalone indicator of conversion.

Bounce Rate and Exit Rate show a strong positive relationship, while purchase sessions are more concentrated at lower rate levels

51.63%

Typical Memory Utilization (Median)

Bounce and Exit Behavior Differ Across Purchase Outcomes



Key observations

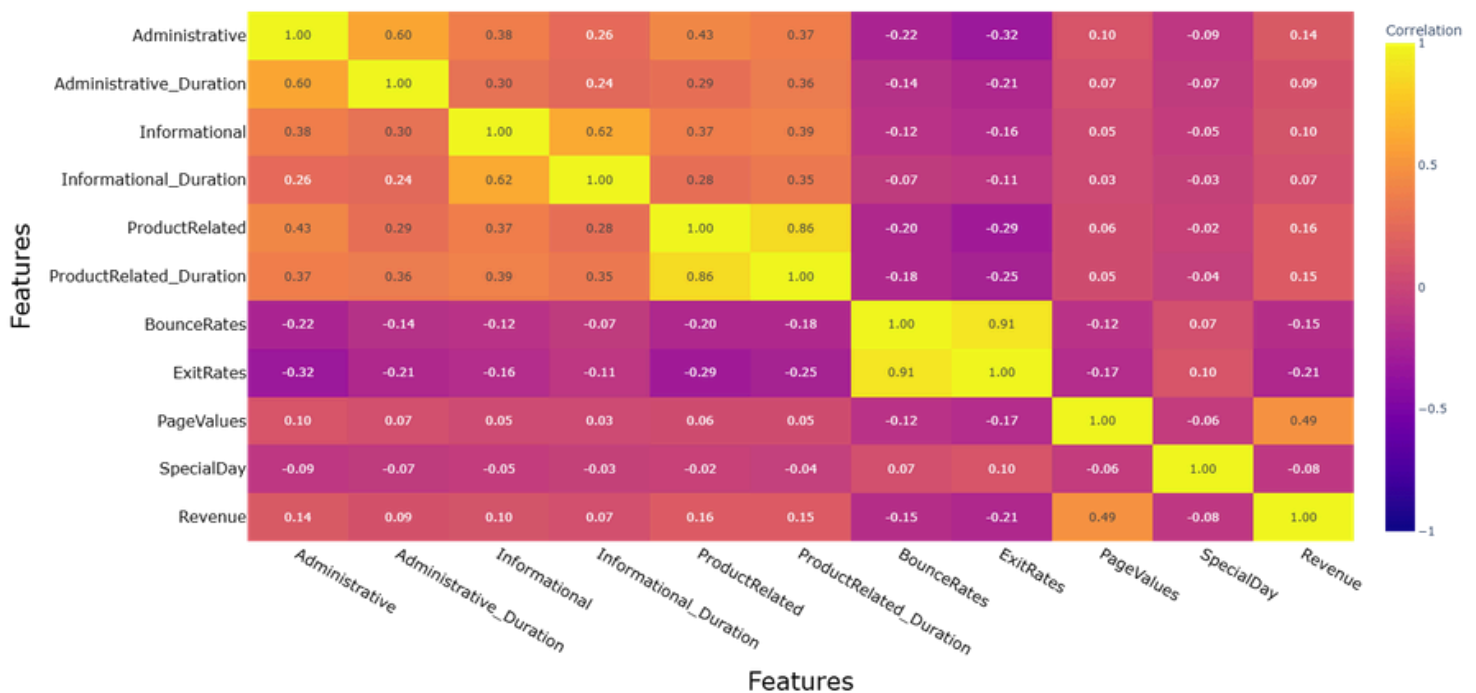
- Bounce Rate and Exit Rate increase together, with most sessions concentrated in the lower range of both measures.
- Purchase sessions are predominantly clustered toward lower Bounce Rate and Exit Rate values, while non-purchase sessions extend more broadly into higher-rate regions.

Business Insights

- Monitor Bounce Rate and Exit Rate together as behavioral indicators when evaluating session quality and conversion performance.
- Investigate sessions with higher bounce and exit rates to identify potential points of friction that may limit progression toward purchase.

PageValues has a strong positive relationship with purchase conversion, while BounceRates ExitRates show negative relationships.

Numerical Features Show Varying Relationships with Purchase Conversion



Key observations

- PageValues has the strongest correlation with Revenue at 0.49, substantially higher than other numerical features.
- ExitRates (-0.21) and BounceRates (-0.15) show negative correlations with Revenue, while ProductRelated (0.16) and Administrative (0.14) show weaker positive relationships.

Business Insights

- Prioritize PageValues as an important behavioral signal when analyzing purchase conversion patterns
- Monitor bounce and exit behavior alongside engagement metrics to identify sessions associated with lower conversion outcomes.



Data Preparation & Feature Engineering



3.1 DATA PREPARATION

Following the exploratory analysis, the dataset was prepared for machine learning.

The preparation process included:

- Selecting the target variable.
- Removing duplicate observations.
- Separating predictor variables from the target.
- Identifying numerical and categorical variables.
- Checking numerical variables for invalid negative values.
- Preparing categorical variables for encoding.
- Creating behavioral engagement features.
- Checking and handling missing values.
- Preparing the dataset for class-imbalanced classification.



3.2 FEATURE ENGINEERING

Several session-level behavioral features were created to provide the models with additional information about browsing activity.

Total Pages Viewed

The following page categories were combined to create this feature:

- Administrative
- Informational
- ProductRelated

Total Browsing Duration

The corresponding duration variables were combined to create:

Total_Browsing_Duration

Product Page Share

The proportion of total page views represented by product-related pages was calculated as: *Product_Page_Share*



Average Duration Per Page

Average browsing duration per viewed page was calculated as:

Avg_Duration_Per_Page

Product Duration Share

The proportion of total browsing duration spent on product-related pages was calculated as: ***Product_Duration_Share***

Undefined ratios caused by zero denominators were replaced with zero.

The final feature-engineered dataset contained:

- **12,205 observations**
- **23 variables**
- **0 missing values**





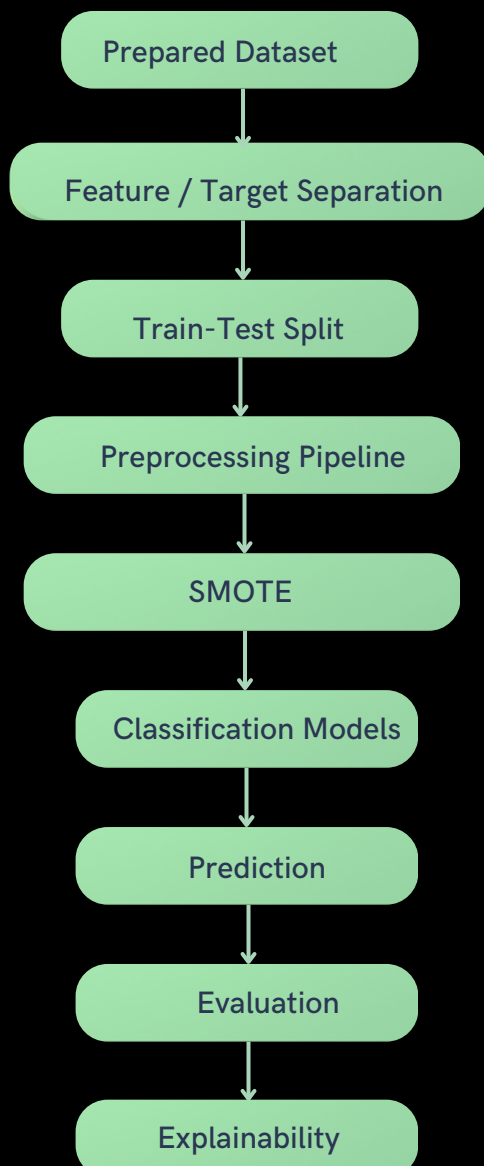
Model Development & Experimentation



4.1 MODELING APPROACH

The problem was formulated as a supervised binary classification problem, with **Revenue** as the target variable.

The modeling workflow consisted of:

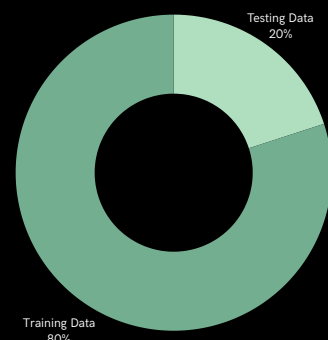


Train-Test Split

A stratified 80/20 train-test split was used to preserve the target distribution.

The resulting datasets contained:

- **Training observations: 9,764**
- **Testing observations: 2,441**



4.2 PREPROCESSING PIPELINE

A unified preprocessing pipeline was used to ensure that numerical and categorical variables were transformed consistently.

Categorical Pipeline



Numerical Pipeline



Binary Features

The ***Weekend*** variable was passed through directly.



4.3 CLASS IMBALANCE HANDLING

The target variable is imbalanced, with purchase sessions representing only approximately 15.63% of the prepared dataset.

To address this imbalance, **SMOTE** (*Synthetic Minority Over-sampling Technique*) was applied to the training data.

SMOTE was incorporated within the machine learning pipeline so that synthetic samples were generated only during training and cross-validation, while the original test set remained unchanged.



4.4 CLASSIFICATION MODELING

Multiple classification approaches were developed and evaluated rather than relying on a single algorithm.

The models included:

- Logistic Regression
- Ridge Classifier
- Decision Tree
- Random Forest
- Tuned Random Forest
- Extra Trees
- Tuned Extra Trees
- Gradient Boosting
- Tuned Gradient Boosting
- HistGradientBoosting
- Tuned HistGradientBoosting
- XGBoost
- Tuned XGBoost
- Voting Classifier
- Stacking Classifier

Hyperparameter tuning was applied to selected tree-based models using cross-validation.

MLflow was also used to track model experiments and parameters.





Model Evaluation & Comparison



5.1 EVALUATION METRICS

The classification models were evaluated using:

Accuracy

Accuracy measures the overall proportion of correctly classified sessions.

Precision

Precision measures the proportion of sessions predicted as purchases that were actually purchases.

Recall

Recall measures the proportion of actual purchase sessions correctly identified by the model.

F1-Score

F1-score provides a balance between precision and recall and was used as the primary model comparison metric because the purchase class is the minority class.

ROC-AUC

ROC-AUC measures the model's ability to distinguish between purchase and non-purchase sessions across classification thresholds.



5.2 MODEL COMPARISON

The final test-set results were:

Feature	Accuracy	Precisiom	Recall	F1
Gradient Boosting	0.8984	0.6558	0.7382	0.6946
Tuned Gradient Boosting	0.8935	0.6326	0.7618	0.6912
Tuned XGBoost	0.8943	0.6390	0.7461	0.6884
Tuned Random Forest	0.8976	0.6594	0.7147	0.6859
Tuned HistGradientBoosting	0.8914	0.6303	0.7408	0.6811
Voting Classifier	0.8906	0.6275	0.7408	0.6795
Random Forest	0.9009	0.6882	0.6702	0.6790
Stacking Classifier	0.8964	0.6617	0.6911	0.6761
HistGradientBoosting	0.8959	0.6641	0.6780	0.6710
XGBoost	0.8943	0.6667	0.6492	0.6578
Logistic Regression	0.8460	0.5050	0.7932	0.6171
Tuned Extra Trees	0.8726	0.6053	0.5340	0.5675
Decision Tree	0.8558	0.5350	0.5995	0.5654
Ridge Classifier	0.7833	0.5654	0.7723	0.5273
Extra Trees	0.8771	0.6640	0.4346	0.5253



5.3 FINAL MODEL

Gradient Boosting was selected as the final model based on its test-set F1-score.

Its performance was:

- Accuracy: 89.84%
- Precision: 65.58%
- Recall: 73.82%
- F1-score: 69.46%
- ROC-AUC: 93.20%

The tuned HistGradientBoosting model achieved the highest ROC-AUC at 93.64%, but its F1-score was lower at 68.11%.

Therefore, F1-score was retained as the primary metric for final model selection because the project focuses on identifying purchase sessions within an imbalanced target.

5.4 CONFUSION MATRIX

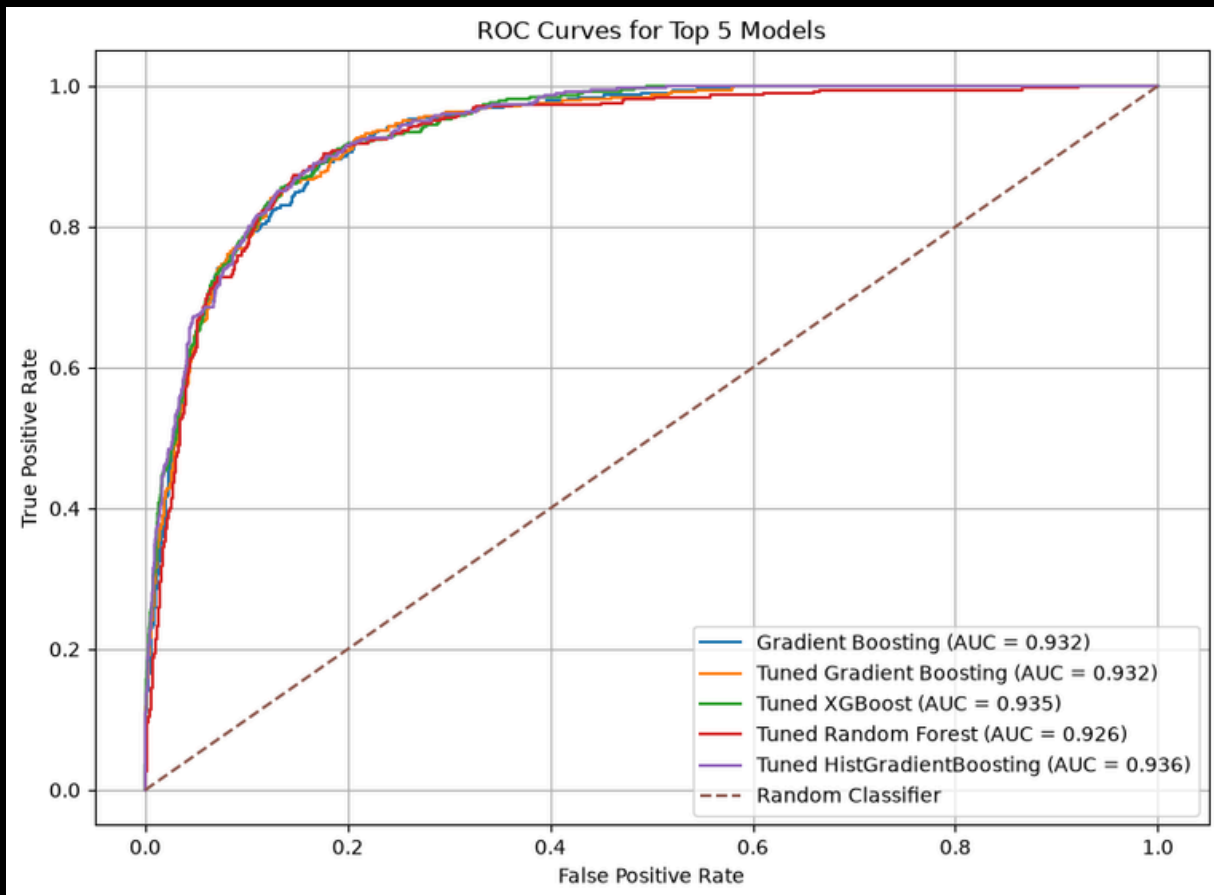
	Actual No Purcha	Actual Purchase
Actual No Purchase	1,911	148
Actual Purchase	100	282

The model correctly classified 2,193 of 2,441 test sessions.

The model produced: 148 False Positives & 100 False Negatives



5.5 ROC-AUC ANALYSIS



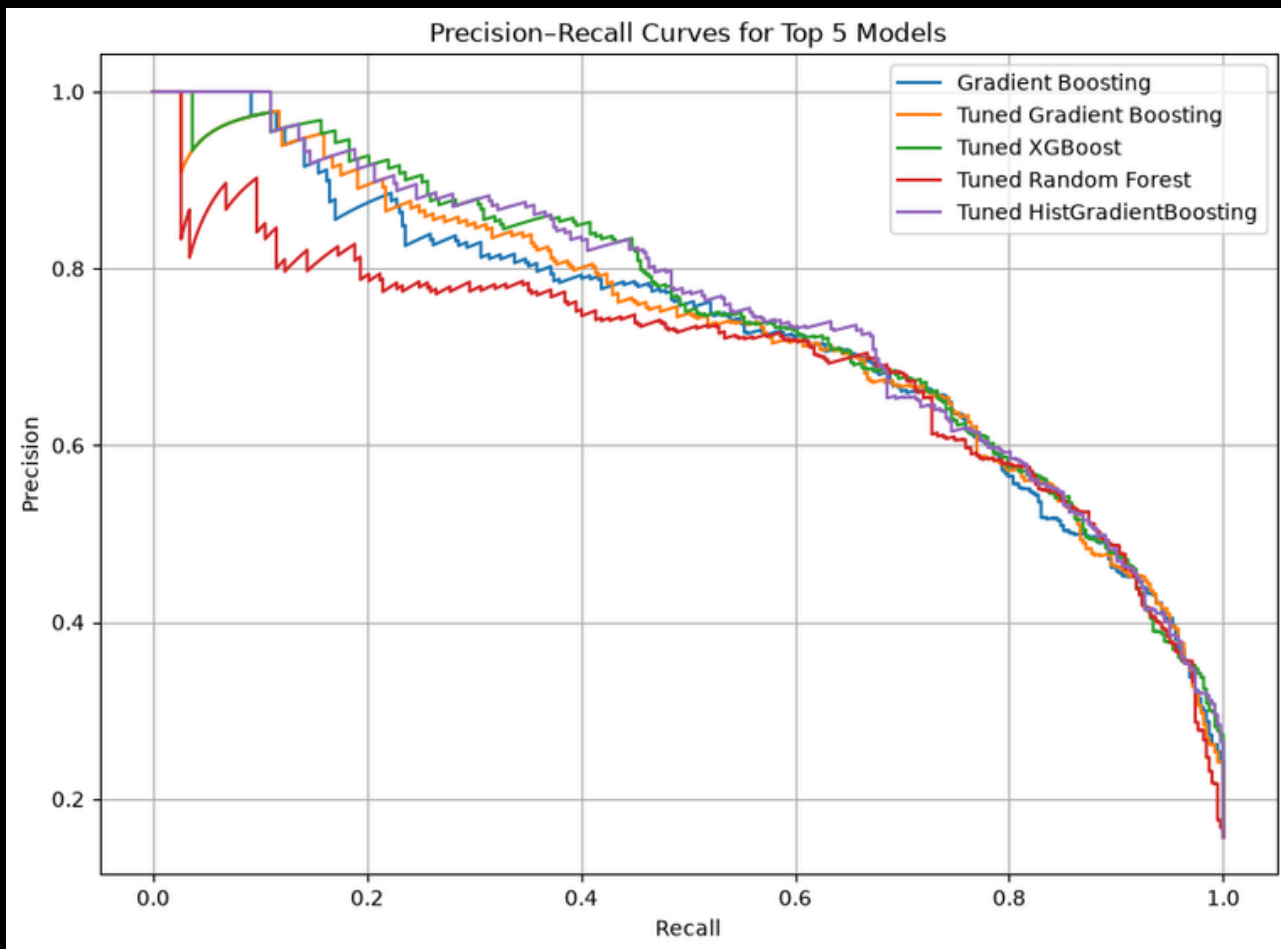
The final Gradient Boosting model achieved a ROC-AUC of 0.931974.

The highest ROC-AUC among the evaluated models was achieved by Tuned HistGradientBoosting at 0.936369.

This indicates strong separation between purchase and non-purchase sessions across classification thresholds.



5.6 PRECISION-RECALL ANALYSIS



Precision-Recall analysis was performed because the purchase class represents a minority of sessions.

The analysis provides additional information about the trade-off between identifying purchase sessions and limiting false purchase predictions.



5.7 CLASSIFICATION THRESHOLD ANALYSIS

The Gradient Boosting model was evaluated across thresholds from 0.10 to 0.90.

The highest F1-score among the tested thresholds was achieved at the default 0.50

- Precision: 65.58%
- Recall: 73.82%
- F1-score: 69.46%

Threshold	MAE	RMSE	F1
0.30	55.18%	80.89%	65.61%
0.40	60.53%	77.49%	67.97%
0.50	65.58%	73.82%	69.46%
0.60	67.09%	68.85%	67.96%
0.70	70.89%	64.40%	67.49%
0.80	74.72%	52.62%	61.75%

The default threshold therefore provided the strongest F1-score among the tested values, so no custom threshold was introduced.



5.8 PREDICTION ERROR ANALYSIS

Prediction errors were investigated using the final model.

The test-set error distribution was:

- Correct predictions: 2,193
- False positives: 148
- False negatives: 100

False Negatives

False negatives represent actual purchase sessions incorrectly classified as non-purchases.

Their median values included:

- PageValues: 0
- ProductRelated: 35.5
- ProductRelated_Duration: 1,413.51
- Total_Pages_Viewed: 40.5
- Total_Browsing_Duration: 1,613.47
- Prediction probability: 0.226



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False Positives

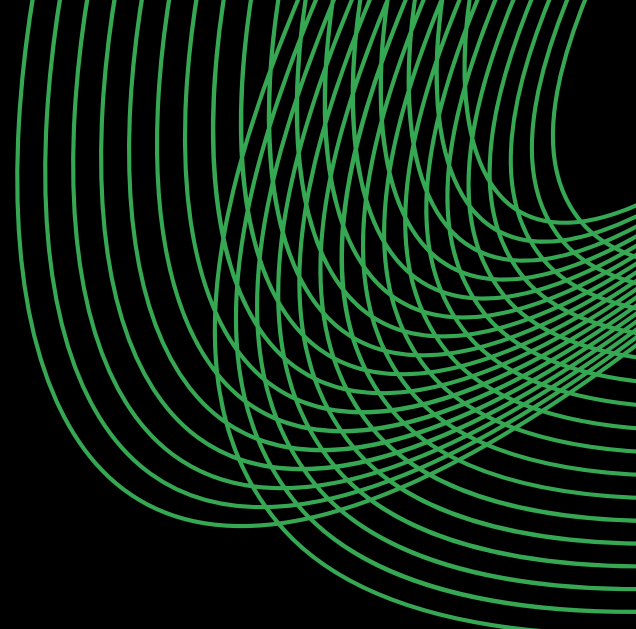
False positives represent non-purchase sessions incorrectly classified as purchases.

Their median values included:

- PageValues: 17.43
- ProductRelated: 36
- ProductRelated_Duration: 1,344.38
- Total_Pages_Viewed: 42
- Total_Browsing_Duration: 1,541.94
- Prediction probability: 0.781

The error analysis shows that false positives and false negatives can exhibit similar levels of browsing engagement, making some sessions difficult to distinguish using behavioral activity alone.





Explainability & Final Model



6.1 WHY EXPLAINABILITY WAS INCLUDED

Accurate predictions alone are not sufficient for understanding e-commerce conversion behavior.

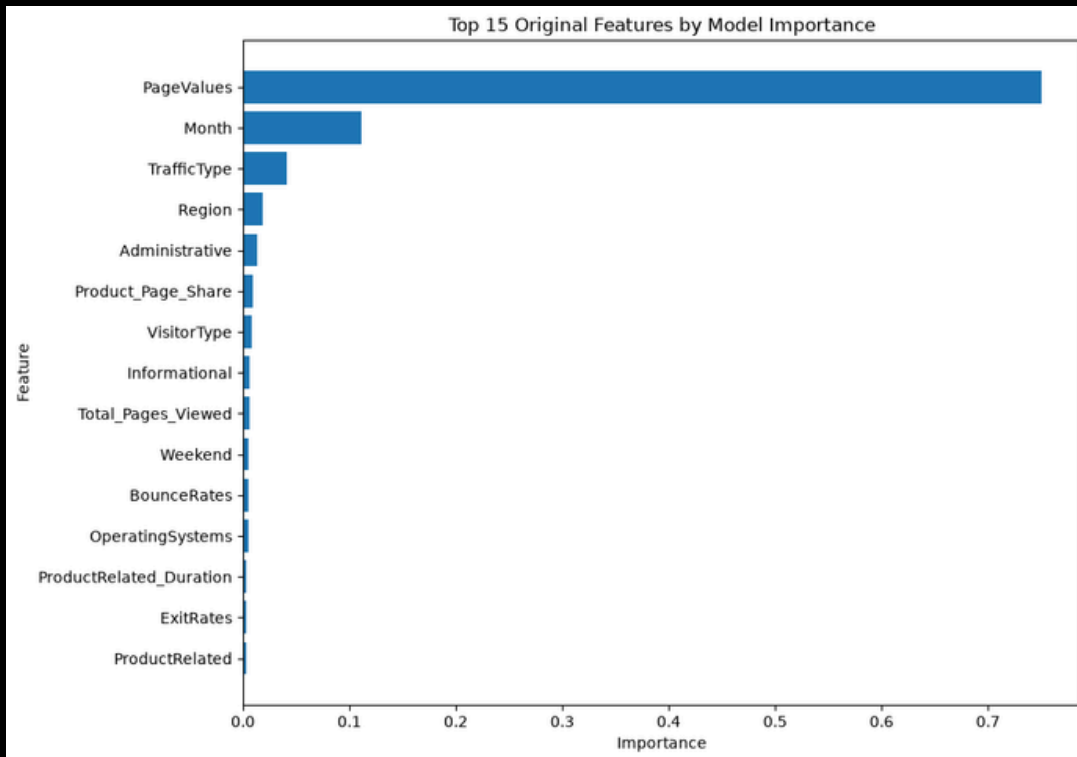
The explainability analysis was therefore used to understand:

- Which variables influence model predictions?
- Which behavioral signals are most important?
- How do encoded categorical variables contribute?
- Where does the model make incorrect predictions?

Feature importance and permutation importance were used to investigate model behavior.



6.2 FEATURE IMPORTANCE



The highest-ranked model features were:

Rank	Feature	Importance
1	PageValues	74.99%
2	Month - November	9.83%
3	TrafficType - 2	2.93%
4	Administrative	1.35%
5	Administrative	0.95%
6	Region - 1	0.90%
7	Informational	0.66%
8	Month - May	0.65%
9	Total_Pages_Viewed	0.59%
10	Weekend	



6.3 ORIGINAL FEATURE IMPORTANCE

Because categorical variables were one-hot encoded, individual encoded categories can make interpretation difficult.

Therefore, encoded feature importance was aggregated back to the original feature groups. The resulting ranking included:

Rank	Feature	Importance
1	PageValues	74.99%
2	Month	11.17%
3	TrafficType	4.15%
4	Region	1.85%
5	Administrative	1.35%
6	Product_Page_Share	0.95%
7	VisitorType	0.82%
8	Informational	0.66%
9	Total_Pages_Viewed	0.59%
10	Weekend	0.55%

PageValues dominates the model's learned feature importance, accounting for approximately 75% of total model importance.

Month and TrafficType also contribute noticeable importance, while the remaining features have substantially smaller individual contributions.



6.5 ERROR PROFILE ANALYSIS

The error analysis provides additional insight into the types of sessions that are difficult to classify.

False-negative sessions had:

- Median PageValues of 0
- Median ProductRelated pages of 35.5
- Median ProductRelated duration of approximately 1,413.51
- Median prediction probability of 0.226

False-positive sessions had:

- Median PageValues of 17.43
- Median ProductRelated pages of 36
- Median ProductRelated duration of approximately 1,344.38
- Median prediction probability of 0.781

The similarity in browsing engagement between the two error groups indicates that high session activity alone does not always distinguish purchase from non-purchase behavior.



6.6 EXPLAINABILITY SUMMARY

The explainability analysis demonstrates two important aspects of the final model.

Strength

The model identifies PageValues, Month, TrafficType, Region, and several behavioral features as important signals for distinguishing purchase and non-purchase sessions.

Weakness

The model still produces false positives and false negatives among sessions with similar browsing and product-engagement characteristics.

This indicates that some purchase decisions cannot be completely separated using the available session-level variables.

